

Optimization Evaluated Practical Session

03/18/26 - 3 Hours

Abstract

The aim of this practical session is to write the algorithms that we have studied during the class. I will ask you to study the convexity of the function, find the global minimum or local minima. We will also see the influence of the learning rate for the gradient descent with a fixed step and see a condition of convergence of this algorithm. Provide a python notebook containing answers to the questions, with markdown cells when needed for comments and remarks. Each question in teal should be done by hand, you should write the solution and derivations using latex (doable in a markdown cell in python notebook)

Introduction

We will use the following functions :

$$f_1(x, y) = x^2 + \frac{y^2}{20},$$

$$f_2(x, y) = \frac{x^2}{2} + \frac{y^2}{2},$$

$$f_3(x, y) = (1 - x)^2 + 10(y - x^2)^2,$$

$$f_4(x, y) = \frac{x^2}{2} + x \cos(y).$$

For the function f_3 we will take $(x_0, y_0) = (-1, 1)$ as the initialization of our algorithms.

1. Compute the gradient of each functions
2. Which of the functions are convex ? Why ? You can use what we have studied during the lessons or in exercises.
3. Plot these functions.
4. What is the global minimum of each functions ?

We now want to solve the following optimization problem

$$\min_{(x,y) \in \mathbb{R}^2} f(x, y),$$

for each function f using the different algorithm studied in class.

1 Gradient descent with a constant learning rate

We recall that the gradient descent algorithm with constant learning rate $\eta > 0$ updates the weights at each iteration as follows :

$$u_{k+1} \leftarrow u_k - \eta \nabla f(u_k).$$

1. Define a function called *gradientdescent* using this gradient descent algorithm.
2. Use your function for the different values of η with the function (f_1, f_2, f_3, f_4) . What do you notice ? Represent the convergence of (x, y) in a graph (evolution of parameters over time)

3. Now we consider the function f_2 , if you have not done it, test the algorithm for $\eta = 1.9$ and $\eta = 2.1$. Give a condition on η so that the algorithm converges.

We suppose that the function f is α -elliptical and the gradient function ∇f is L -lipschitzien. We can show that this algorithm converges if we take η such that : $0 < \eta < \frac{2\alpha}{L^2}$.

- (a) Find the value of L such that :

$$\|\nabla f(x_1, y_1) - \nabla f(x_2, y_2)\| \leq L\|(x_1, y_1) - (x_2, y_2)\|$$

- (b) Compute λ_{min} the smallest eigenvalue of f . We admit that $\alpha = \lambda_{min}$ and conclude.
(c) Try to do the same for the function f_1 .

2 Gradient descent with optimal step

Now the learning rate is no more constant, it is determined by solving the problem :

$$\eta^{(k)} = \underset{\eta > 0}{\operatorname{Argmin}} f(u_k - \eta \nabla f(u_k)).$$

1. Give an explicit expression of η for the first and/or second function(s).
2. Implement the algorithm.
3. Solve the problem of minimization of function f_1 and compare to the previous algorithm.
4. Do the same for the function f_3 .

3 Newton's Method (30 minutes)

The Newton's Method is solving Euler's Equation

$$\nabla f(u) = 0.$$

An iteration of the Newton's algorithm is given by :

$$u_{k+1} \leftarrow u_k - (H_f(u))^{-1} \nabla f(u),$$

where H_f refers to the Hessian matrix of the function f .

Remark : It is possible to improve this method combining it with a line search algorithm, setting :

$$u_{k+1} \leftarrow u_k - \eta (H_f(u))^{-1} \nabla f(u),$$

where η is constant or satisfies the Wolfe's condition.

1. Compute the Hessian matrix for the functions f_1 , f_3 and f_4 .
2. Implement the Newton's method for the function f_3 and f_4 to find the global minimum.
3. For the function f_2 and/or f_3 , compare the convergence of the three algorithms.
4. What are the advantage(s) and drawback(s) of this method ?